

Logistic
Regression

Classification

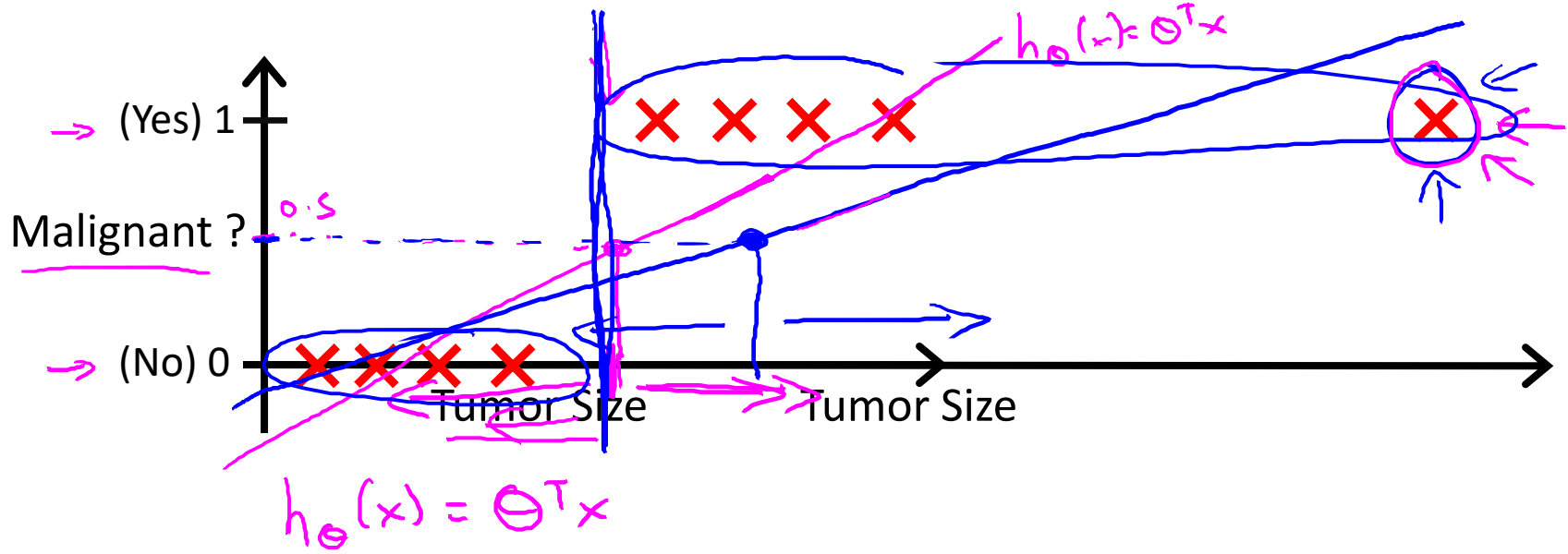
Classification

- Email: Spam / Not Spam?
- Online Transactions: Fraudulent (Yes / No)?
- Tumor: Malignant / Benign ?

→ $y \in \{0, 1\}$

0: "Negative Class" (e.g., benign tumor)
1: "Positive Class" (e.g., malignant tumor)

→ $y \in \{0, 1, 2, 3\}$



→ Threshold classifier output $h_{\theta}(x)$ at 0.5:

→ If $h_{\theta}(x) \geq 0.5$, predict "y = 1"

If $h_{\theta}(x) < 0.5$, predict "y = 0"

Classification:

$$y = 0 \text{ or } 1$$

$h_{\theta}(x)$ can be > 1 or < 0

Logistic Regression: $0 \leq h_{\theta}(x) \leq 1$

Classification

Logistic Regression

Hypothesis
Representation

Logistic Regression Model

Want $0 \leq h_{\theta}(x) \leq 1$

$$h_{\theta}(x) = g(\theta^T x)$$

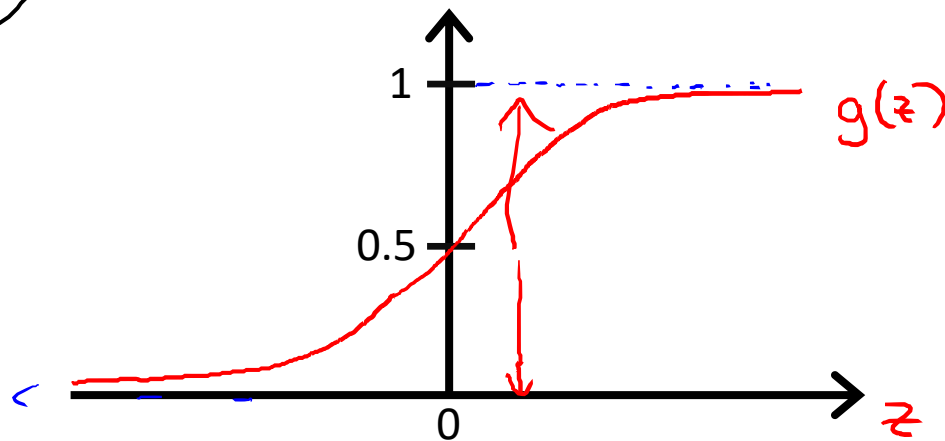
$$\rightarrow g(z) = \frac{1}{1 + e^{-z}}$$

$\theta^T x$

Sigmoid function

Logistic function

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$



Parameters θ .

Interpretation of Hypothesis Output

$$h_{\theta}(x)$$

$h_{\theta}(x)$ = estimated probability that $y = 1$ on input x ←

Example: If $\underline{x} = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \leftarrow \\ \text{tumorSize} \end{bmatrix} \leftarrow$

$h_{\theta}(x)$ = 0.7

$y = 1$

Tell patient that 70% chance of tumor being malignant

$$\underline{h_{\theta}(x)} = \underline{P(y=1|x;\theta)}$$

$y = 0 \text{ or } 1$

“probability that $y = 1$, given x ,
parameterized by θ ”

→ $P(y = 0 | \text{ ~~$x$~~ ; ~~$\theta$~~) + \boxed{P(y = 1 | \text{ ~~x~~ ; ~~θ~~)}$ = 1

→ $P(y = 0 | x; \theta) = 1 - \boxed{P(y = 1 | x; \theta)}$

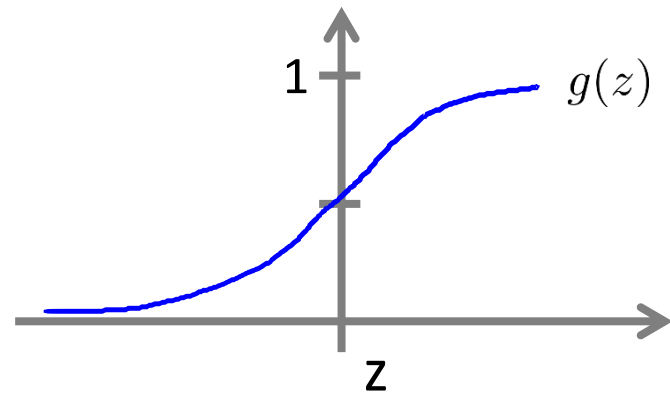
Logistic Regression

Decision boundary

Logistic regression

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1+e^{-z}}$$



Suppose predict “ $y = 1$ ” if $h_{\theta}(x) \geq 0.5$

$$\theta^T x \geq 0$$

$$g(z) \geq 0.5 \\ \text{when } z \geq 0$$

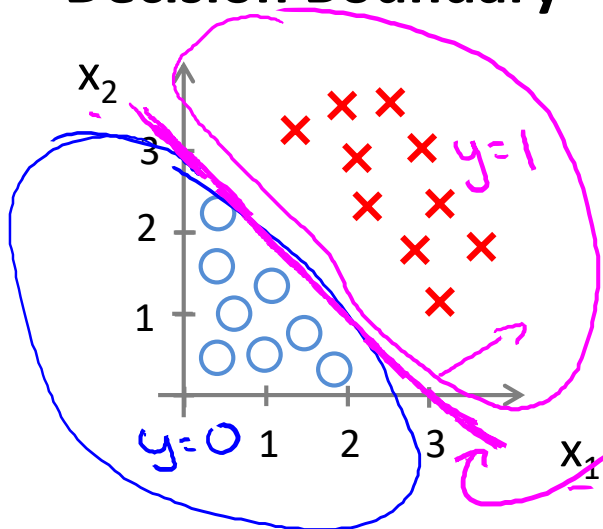
$$h_{\theta}(x) = g(\theta^T x)$$

predict “ $y = 0$ ” if $h_{\theta}(x) < 0.5$

$$\theta^T x < 0$$

$$g(z) < 0.5 \\ \text{when } z < 0$$

Decision Boundary



$$\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \leftarrow$$

$$\rightarrow h_{\theta}(x) = g(\underbrace{\theta_0}_{-3} + \underbrace{\theta_1}_{1}x_1 + \underbrace{\theta_2}_{1}x_2)$$

Decision boundary

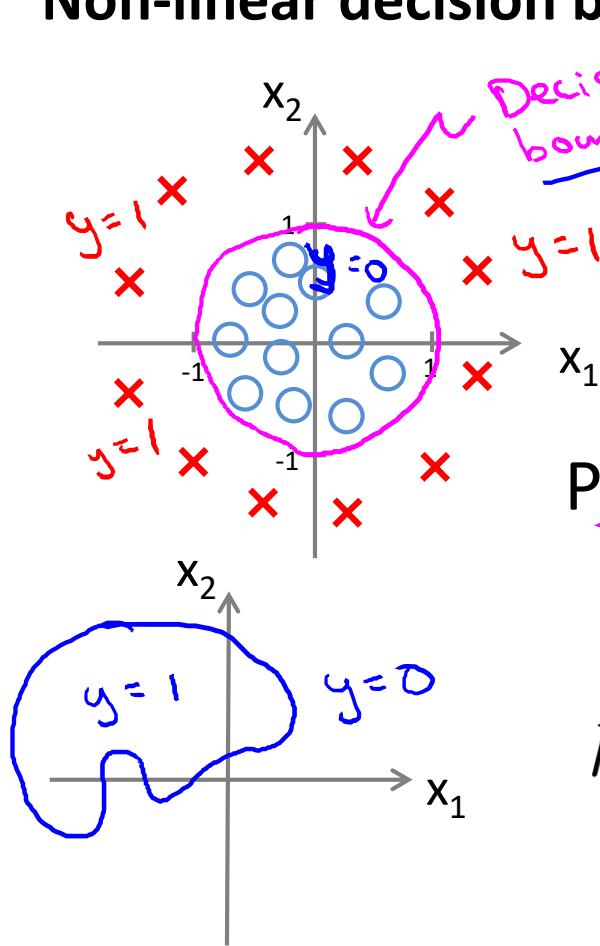
Predict " $y = 1$ " if $\underbrace{-3 + x_1 + x_2}_{\theta^T x} \geq 0$

x_1, x_2
 $\rightarrow h_{\theta}(x) = 0.5$
 $x_1 + x_2 = 3$

$\rightarrow \underline{x_1 + x_2 \geq 3}$

$\rightarrow x_1 + x_2 < 3$
 $y = 0$

Non-linear decision boundaries



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

$\begin{matrix} \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} \end{matrix}$

$\downarrow \theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Predict " $y = 1$ " if $-1 + x_1^2 + x_2^2 \geq 0$

$\boxed{x_1^2 + x_2^2 = 1}$

$\underbrace{-1 + x_1^2 + x_2^2}_{x_1^2 + x_2^2 \geq 1} \geq 0$

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 \underline{x_1^2} + \theta_4 \underline{x_1^2 x_2} + \theta_5 \underline{x_1^2 x_2^2} + \theta_6 \underline{x_1^3 x_2} + \dots)$$

Logistic Regression

Cost function

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

m examples

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix} \quad \mathbb{R}^{n+1}$$

$$\underline{x_0 = 1}, \underline{y \in \{0, 1\}}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

How to choose parameters θ ?

Cost function

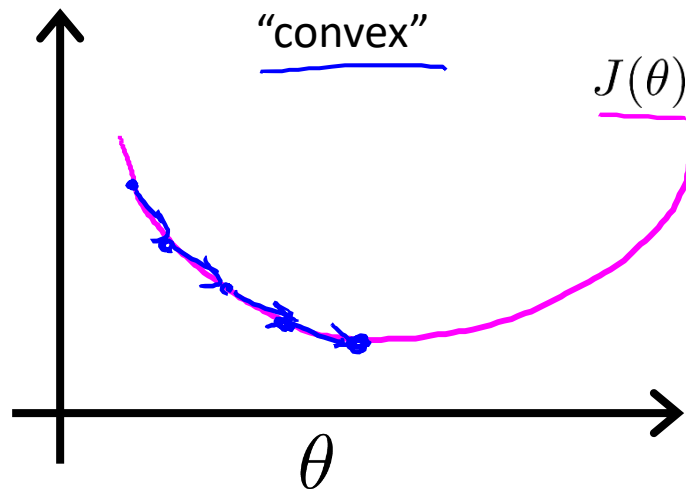
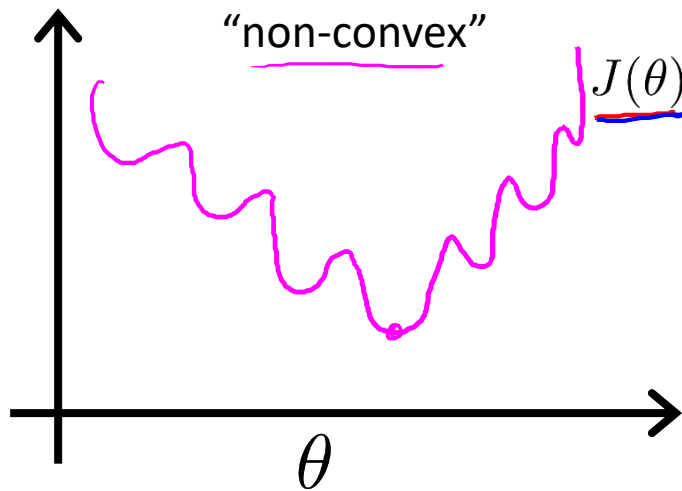
→ ~~Linear~~ regression:
logistic

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$\rightarrow \text{cost}(h_{\theta}(x^{(i)}), y)$

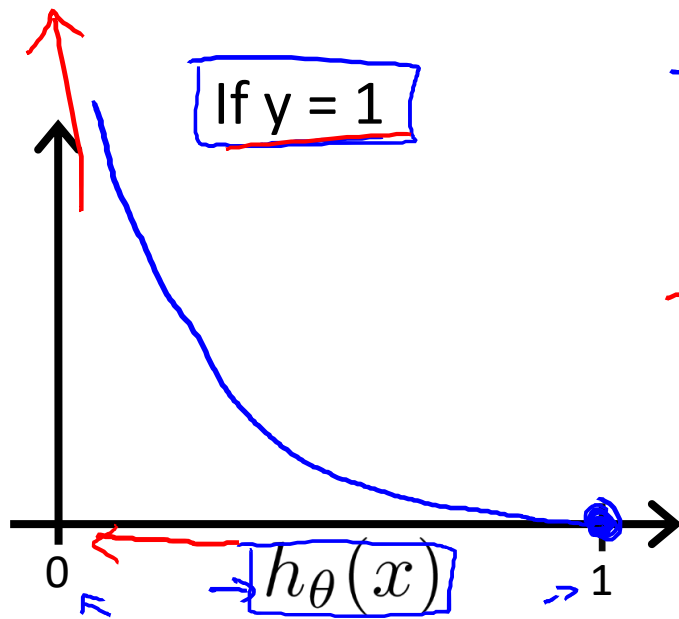
$$\text{Cost}(h_{\theta}(x), y) = \frac{1}{2} (h_{\theta}(x) - y)^2$$

$$\frac{1}{1 + e^{-\theta^T x}}$$



Logistic regression cost function

$$\text{Cost}(\underline{h_\theta(x)}, y) = \begin{cases} \boxed{-\log(h_\theta(x))} & \text{if } y = 1 \\ \underline{-\log(1 - h_\theta(x))} & \text{if } y = 0 \end{cases}$$



→ Cost = 0 if $y = 1, h_\theta(x) = 1$

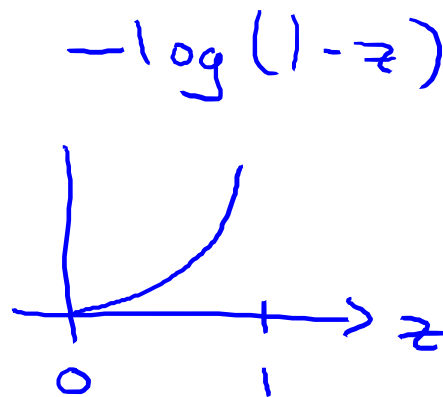
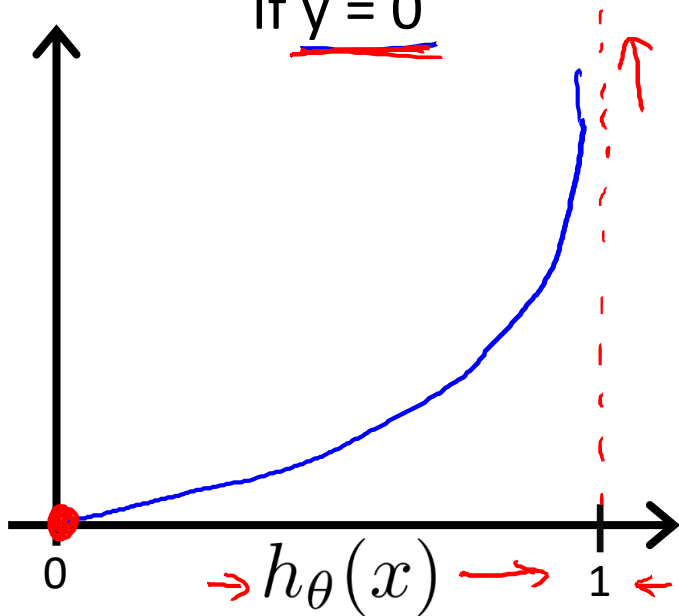
[But as $h_\theta(x) \rightarrow 0$
 $\text{Cost} \rightarrow \infty$

→ Captures intuition that if $h_\theta(x) = 0$,
(predict $P(y = 1|x; \theta) = 0$), but $y = 1$,
we'll penalize learning algorithm by a very
large cost.

Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

If $y = 0$



Logistic Regression

Simplified cost function
and gradient descent

Logistic regression cost function

$$\rightarrow J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$\rightarrow \text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Note: $y = 0$ or 1 always

$$\rightarrow \text{Cost}(h_{\theta}(x), y) = -y \log(h_{\theta}(x)) - (1-y) \log(1 - h_{\theta}(x))$$

If $y=1$: $\text{Cost}(h_{\theta}(x), y) = -\log h_{\theta}(x)$

If $y=0$: $\text{Cost}(h_{\theta}(x), y) = -\log(1 - h_{\theta}(x))$

Logistic regression cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$



To fit parameters θ :

$$\min_{\theta} J(\theta)$$

Get θ

To make a prediction given new x :

$$\text{Output } \underline{h_{\theta}(x)} = \frac{1}{1 + e^{-\theta^T x}}$$

$$\underline{p(y=1 | x; \theta)}$$

Gradient Descent

$$\rightarrow J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

(simultaneously update all θ_j)

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\rightarrow \theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

}

(simultaneously update all θ_j)

$$\Theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix} \leftarrow \text{for } i=0 \text{ to } n$$

$$h_{\theta}(x) = \Theta^T x$$

$$\rightarrow h_{\theta}(x) = \frac{1}{1 + e^{-\Theta^T x}}$$

Algorithm looks identical to linear regression!

Logistic Regression

Multi-class classification:
One-vs-all

Multiclass classification

Email foldering/tagging: Work, Friends, Family, Hobby

$y=1$ $y=2$ $y=3$ $y=4$

Medical diagrams: Not ill, Cold, Flu

$y=1$ 2 3

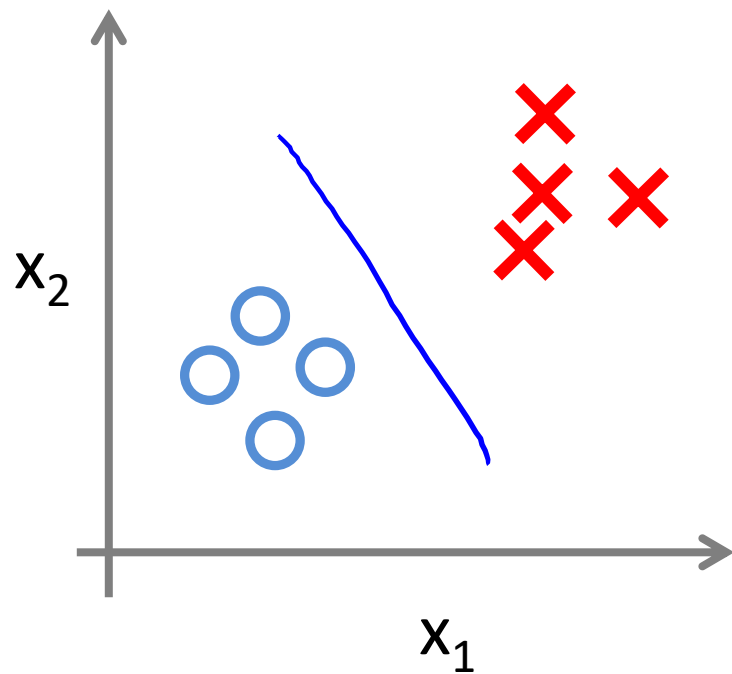
Weather: Sunny, Cloudy, Rain, Snow

$y=1$ 2 3 4 $\leftarrow$

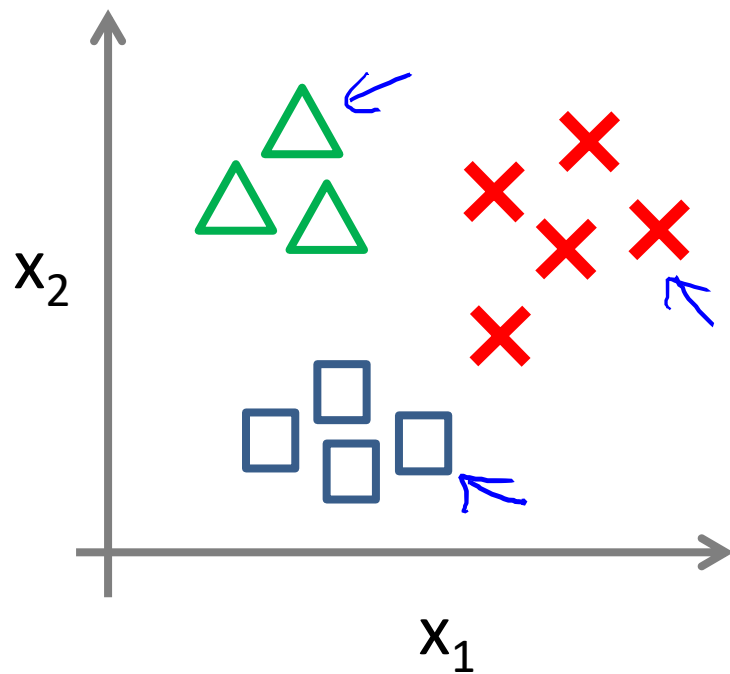


A horizontal line with tick marks at 0, 1, 2, and 3.

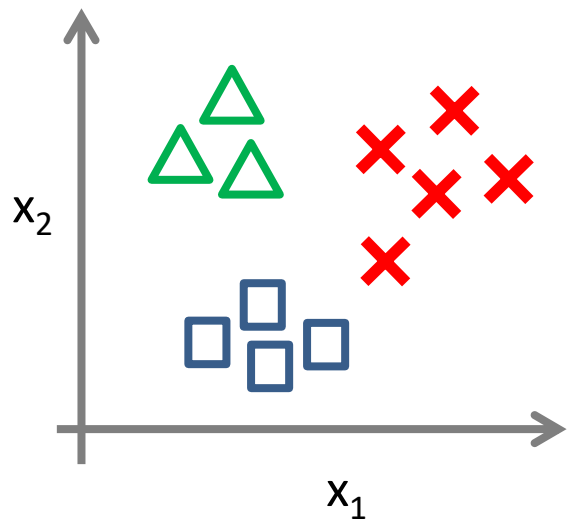
Binary classification:



Multi-class classification:



One-vs-all (one-vs-rest):

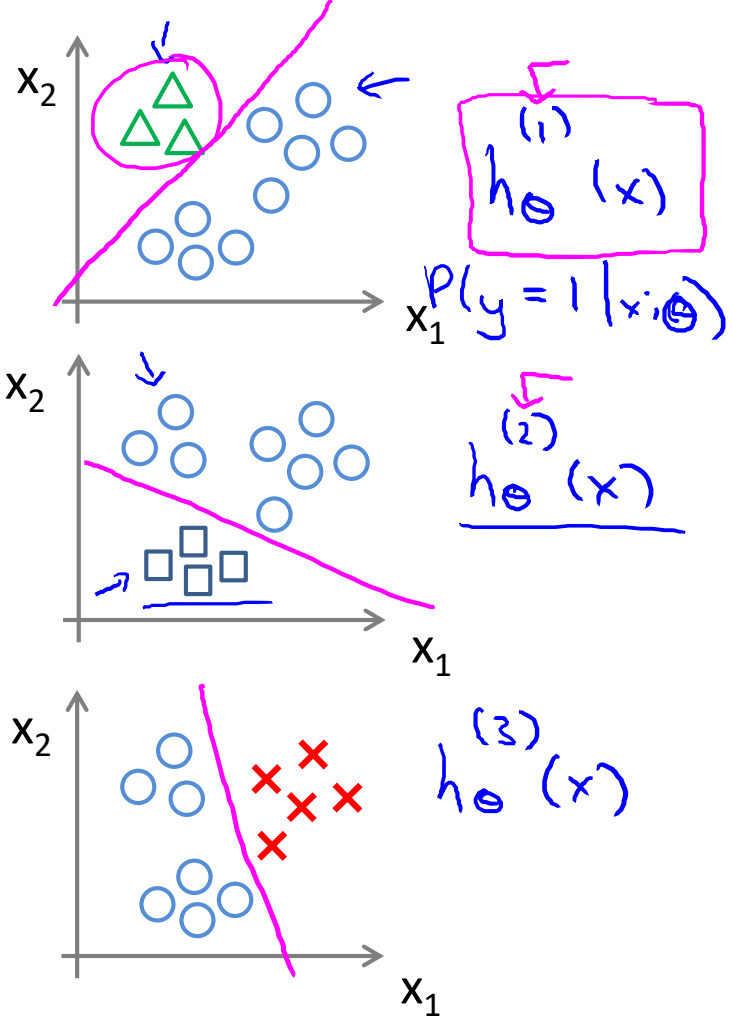


Class 1:  

Class 2:  

Class 3:  

$$h_{\theta}^{(i)}(x) = P(y = i | x; \theta) \quad (i = 1, 2, 3)$$



One-vs-all

Train a logistic regression classifier $\underline{h_{\theta}^{(i)}(x)}$ for each class $\underline{i}$ to predict the probability that $\underline{y = i}$.

On a new input $\underline{x}$, to make a prediction, pick the class i that maximizes

$$\max_{\underline{i}} \underline{h_{\theta}^{(i)}(x)}$$
